

Positive Square Energy of Bridge-Separated Tetracyclic Cacti

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Abstract

Let G be a connected tetracyclic cactus of order n . We prove $s^+(G) > n$ whenever its four cyclic blocks do not all belong to one connected shared-cut cluster. The sharp block-additive doubly nonnegative bound reduces the problem to cycle multisets $\{3, 3, 3, q\}$ and $\{3, 3, 5, 5\}$. We settle every bridge-separated incidence by adaptive induced packetization. Two apparent obstructions disappear only after one allows a hostile cycle to be sacrificed into path fragments: a shared $C_3 - C_5 - C_3$ cluster plus a remote pentagon, and a shared three-triangle cluster plus a remote 1 (mod 4) cycle. Arbitrary connector trees and arbitrary trees attached at arbitrary vertices are allowed. The case in which all four cycles form one shared-cut cluster is not asserted here.

1 Preliminaries

All graphs are finite, simple, and undirected. If the adjacency eigenvalues are $\lambda_1, \dots, \lambda_n$, put

$$s^+(G) = \sum_{\lambda_j > 0} \lambda_j^2, \quad \sigma(G) = s^+(G) - |V(G)|.$$

We use induced-subgraph superadditivity [1]: for every vertex partition $V(G) = V_1 \dot{\cup} \dots \dot{\cup} V_k$,

$$s^+(G) \geq \sum_{j=1}^k s^+(G[V_j]). \quad (1)$$

Join two cyclic blocks in the *shared-cut graph* when they share a cut vertex, and call its connected components shared-cut clusters. In the block-cut tree, contract the incidence subtree belonging to each shared-cut cluster to one marked node; every cluster, including every singleton cyclic block, is marked. Take the minimal subtree spanning these marked nodes and suppress only unmarked degree-two nodes. No marked node is ever suppressed. The resulting tree is the reduced cluster tree. Consequently the interior of every reduced edge contains only bridge blocks and unmarked cut nodes, never an unassigned cyclic block.

Lemma 1.1 (Connector territories). *Let $S_1, \dots, S_k$ be pairwise vertex-disjoint connected subtrees of the reduced cluster tree such that every marked cluster node belongs to exactly one S_i . Then there is a vertex partition into connected induced territories such that the territory corresponding to S_i retains exactly the cyclic blocks in the clusters marked in S_i .*

Proof. Every reduced edge represents a path whose block nodes are all bridge blocks in the block-cut tree; this follows from the preceding convention that every shared-cut cluster, including a singleton cycle, is a nonsuppressed marked node. Assign the internal vertices of each selected connector

path to its selected subtree. Between two selected subtrees, cut one actual bridge on their joining path and assign the two remnants to their respective sides. At an unmarked Steiner node, assign the node and its initial path segments to one selected subtree and cut one bridge on every other incident segment. Every component outside the minimal cyclic hull meets it at a unique vertex, since two attachments would create a cycle in the block-cut tree; assign that entire hanging tree to the territory containing its attachment. The resulting sets are connected. Their induced subgraphs retain precisely the assigned cycles. $\square$

We will also split a cyclic block by assigning a proper path of it to each of two territories. Crossing edges are harmless in (1).

We quote the following previously proved packet estimates [3, 4]. Each permits arbitrary tree attachments; the mixed bound permits arbitrary incidence and connectors, while (5) and (6) have their displayed shared-cut hypotheses. Write $T = C_3$, $P = C_5$, and, for $q \equiv 1 \pmod{4}$,

$$\delta_q = \sec(\pi/q) - 1 < 1, \quad \delta_5 = \sqrt{5} - 2.$$

Then

$$\sigma(T) > 0, \quad \sigma(C_q) \geq -\delta_q, \quad (2)$$

$$\sigma(TT) > 1, \quad \sigma(TC_q) > 1 - \delta_q, \quad (3)$$

$$\sigma(H) \geq 0 \quad \text{for every bicyclic or tricyclic cactus } H, \quad (4)$$

$$\sigma(TTC_q) > 2 - \delta_q \quad \text{if the two triangles share a cut vertex}, \quad (5)$$

$$\sigma(TPP) > 6 - 2\sqrt{5} \quad \text{if the three cycles form one shared-cut cluster}. \quad (6)$$

If every cycle is $3 \pmod{4}$ and the cycle-packing number is at most two, the Sachs half-plane theorem [6] gives $D = s^+ - s^- > 0$. Thus a connected r -cyclic such cactus has

$$\sigma(G) > r - 1. \quad (7)$$

These results are proved in the companion manuscripts on all bicyclic and all tricyclic cacti.

2 The exact residual reduction

For a cactus with bridge-block count b and cycle lengths $\ell_1, \dots, \ell_4$, block counting gives

$$b + \sum_{j=1}^4 \ell_j = n + 3.$$

The sharp cactus DNN constant [5, 7] implies

$$s^+(G) - n \geq 3 - \sum_{j=1}^4 \epsilon_{\ell_j}, \quad (8)$$

where $\epsilon_\ell = 0$ for even ℓ and $\epsilon_\ell = \ell \tan^2(\pi/(2\ell))$ for odd ℓ . Moreover

$$\epsilon_3 = 1, \quad \epsilon_5 = 5 - 2\sqrt{5}, \quad \epsilon_5 + \epsilon_7 < 1,$$

and ϵ_ℓ strictly decreases through odd lengths. Hence (8) proves the theorem unless the cycle multiset is

$$\{3, 3, 3, q\} \quad (q \geq 3), \quad \text{or} \quad \{3, 3, 5, 5\}. \quad (9)$$

3 Three triangles and one further cycle

Proposition 3.1. *Let the cyclic blocks of G have lengths $3, 3, 3, q$, where $q \geq 3$. If the shared-cut graph is disconnected, then $s^+(G) > n$.*

Proof. If q is even, take the shared-cut clusters as induced territories using Lemma 1.1. The territory containing the even cycle is at most tricyclic and has nonnegative surplus by (4) (and is bipartite with zero surplus if it contains no triangle). Since the shared-cut graph is disconnected, another territory contains one, two, or three triangles. Its cycle-packing number is at most two, so its surplus is strictly positive by (7). Every remaining territory has nonnegative surplus. Equation (1) proves the claim.

If $q \equiv 3 \pmod{4}$, partition along the bridge connectors between shared-cut clusters. Every territory contains only favorable cycles and has cycle packing at most two within each nontrivial cluster; singleton territories are favorable unicyclic cacti. Equations (7) and (1) give the result.

Suppose $q \equiv 1 \pmod{4}$, and denote its block by Q . Up to permutation of the triangles, the shared-cut component partitions are

$$T|T|T|Q, \quad TT|T|Q, \quad TQ|T|T, \quad TTT|Q, \quad TT|TQ, \quad TTQ|T.$$

We treat them in turn.

For $T|T|T|Q$, choose a triangle whose marked node has minimum distance from Q in the reduced cluster tree. The unique Q – T path then contains no other marked node. Use this path as one connected subtree and the other two triangle nodes as singleton subtrees; Lemma 1.1 realizes the territories even when the path passes through an unmarked Steiner node. By (2)–(3), the total surplus is strictly greater than $1 - \delta_q > 0$. The same direct accounting settles $TQ|T|T$. The partition $TT|TQ$ has surplus greater than $1 + (1 - \delta_q) > 0$.

For $TT|T|Q$, call the three marked nodes $A = TT$, $B = T$, and $C = Q$. If the B – C path avoids A , Lemma 1.1 gives a TQ packet and a separate TT packet. Otherwise A lies on that path; the connected A – C subtree gives a TTQ packet separated from B . The resulting TTQ territory retains the shared pair of triangles, so (5), together with the remaining triangular packet, gives total surplus greater than $2 - \delta_q > 0$.

For $TTQ|T$, the tricyclic territory has nonnegative surplus by (4), and the singleton triangle has strictly positive surplus.

It remains to treat $TTT|Q$. Let $w \in V(Q)$ be the vertex through which the block-tree route from Q to the triangular cluster leaves Q . Choose $v \in V(Q) \setminus \{w\}$. Let F consist of v together with every component of $G - E(Q)$ attached to Q only at v , and put all other vertices into H . The graph $Q - v$ is a path containing w , so H is connected; its only cycles are the three triangles in one shared-cut cluster. Thus its cycle-packing number is at most two and

$$s^+(H) > |H| + 2$$

by (7). Since F is a tree, $s^+(F) = |F| - 1$. Therefore

$$s^+(G) \geq s^+(H) + s^+(F) > |H| + 2 + |F| - 1 = n + 1.$$

All component partitions have now been covered. □

4 Two triangles and two pentagons

The only delicate packetization is recorded separately.

Lemma 4.1 (Middle-pentagon sacrifice). *Suppose one shared-cut cluster consists of a pentagon P and two disjoint triangles T_1, T_2 meeting P at distinct vertices x_1, x_2 . Suppose a second pentagon Q is bridge-separated from this cluster. Then the vertices can be partitioned into connected induced territories which are either*

1. *a TT bicyclic cactus and a pentagonal unicyclic cactus, or*
2. *a triangular unicyclic cactus and a TP bicyclic cactus.*

In either case $s^+(G) > n$.

Proof. Let z be the first vertex of the shared cluster met by the unique block-tree route from Q . Include Q , its entire connector, and all branches hanging from that connector on the Q side.

If $z \in V(P) \setminus \{x_1, x_2\}$, put z with Q and put $P - z$ together with both triangles on the other side. The latter is connected because $P - z$ is a path containing x_1, x_2 . The two induced territories have cycle multisets $\{Q\}$ and $\{T_1, T_2\}$, respectively. Hence their surplus sum is strictly greater than $-\delta_5 + 1 = 3 - \sqrt{5} > 0$.

Otherwise z lies on one triangle, say T_1 ; this includes $z = x_1$ (which also lies on P) and a connector entering through any branch rooted on T_1 . Put all of T_1 , the connector, and Q on one side. Put $P - x_1$ and T_2 on the other side. The first induced territory is a mixed TP bicyclic cactus; the second is triangular unicyclic because $P - x_1$ is a path containing x_2 . Their surplus sum is strictly greater than $1 - \delta_5 > 0$. The case in which the route enters through T_2 is symmetric.

Every remaining bridge-tree component meets this retained skeleton at only one vertex; otherwise the block-cut graph would contain a cycle. Assign it wholly to the territory containing that attachment. Thus both territories remain connected and induced. In all cases the middle pentagon is deliberately split into proper path fragments, so it creates no uncounted cycle. $\square$

Proposition 4.2. *Let the cyclic blocks of G have lengths 3, 3, 5, 5. If the shared-cut graph is disconnected, then $s^+(G) > n$.*

Proof. Write again $T = C_3$ and $P = C_5$. Up to equal-cycle permutation, the possible shared-cut component partitions are

$$TTP|P, \quad TPP|T, \quad TT|PP, \quad TP|TP, \quad TT|P|P, \quad PP|T|T, \quad TP|T|P, \quad T|T|P|P.$$

For $TPP|T$, use (6) and the strict triangular surplus. For $TT|PP$ and $TP|TP$, the packet sums are respectively greater than 1 and $2(1 - \delta_5)$. For the three-cluster partitions, direct cluster territories give

$$\begin{aligned} TT|P|P : & \quad \sigma(G) > 1 - 2\delta_5 = 5 - 2\sqrt{5} > 0, \\ PP|T|T : & \quad \sigma(G) > 0, \\ TP|T|P : & \quad \sigma(G) > 1 - 2\delta_5 > 0. \end{aligned}$$

For $T|T|P|P$, consider the reduced tree spanning the four cycle nodes. If a triangle is a leaf, cut its first bridge: the triangular packet has positive surplus and the remaining tricyclic cactus has nonnegative surplus. If neither triangle is a leaf, the only leaves are the two pentagons. A finite tree with exactly two leaves is a path, so the cycle order is $P - T - T - P$. Cut between the triangles to obtain two induced TP bicyclic territories, whose total surplus is greater than $2(1 - \delta_5) > 0$.

Finally consider $TTP|P$. If the two triangles in the triple cluster share a cut vertex, (5) gives total surplus greater than $2 - 2\delta_5 > 0$. Otherwise the shared-cut incidence tree is necessarily $T - P - T$, with the pentagon in the middle and distinct triangle cuts. This is exactly Lemma 4.1. The enumeration is exhaustive. $\square$

5 Main theorem

Theorem 5.1. *Let G be a connected tetracyclic cactus on n vertices. If its four cyclic blocks do not all belong to one connected shared-cut cluster, then*

$$\boxed{s^+(G) > n}.$$

Proof. The DNN estimate (8) settles every cycle multiset except (9). Proposition 3.1 settles the first residual family and Proposition 4.2 settles the second. Every construction uses vertex-disjoint induced territories, so arbitrary crossing bridge edges are discarded only through the valid super-additivity inequality (1); no edge-monotonicity is used. $\square$

6 Scope

The theorem does not cover a tetracyclic cactus in which all four cycles form one connected shared-cut cluster. In particular, it does not claim the full tetracyclic-cactus case. An exact repository census finds twenty quotient-isomorphism types for the shared $\{3, 3, 5, 5\}$ residual, and the direct coefficientwise two-pentagon phase certificate succeeds for only one; see `research/c3-c3-c5-c5-shared-cluster-census`. The other cases remain unresolved by that certificate.

References

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- [3] Companion manuscript, *Positive Square Energy of Every Bicyclic Cactus*, repository path `all-bicyclic-cacti/paper.tex`.
- [4] Companion manuscript, *Positive Square Energy of Every Tricyclic Cactus*, repository path `all-tricyclic-cacti/paper.tex`.
- [5] Companion manuscript, *The Optimized Liu–Tang–Zhang Constant for Cactus Graphs*, repository path `sharp-cactus-dnn/paper.tex`.
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